

4EB00 Q1 2026

Special Topic: The Jet Engine

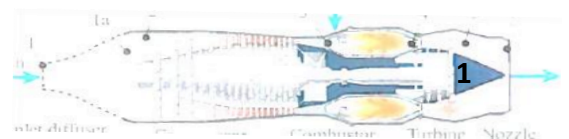
Groep **10**

	Naam	Studentnr
1		
2		
3		
4		

Assignment Conditions

Group 10 settings

Name	Value	Unit
Fuel	H2	
Tamb	300	[K]
P3overP2	9	[-]
Pamb	100000	[Pa]
mfurate	0.58	[kg/s]
AF	204.42	[kg/kg]
v1	200	[m/s]



Assignment Cycle Analysis of the Jet Engine based on NASA tables

The Group 10 ideal-cycle model predicts an exhaust velocity of 778.83 m/s. Table 1 gives the six calculated states. The following sections explain the conservation equations and include numbered excerpts from JetEngine_Group10.m. All line numbers refer to the script included in the submitted scripts ZIP.

State	1	2	3	4	5	6
P (kPa)	100.00	125.11	1125.98	1125.98	423.74	100.00
T (K)	300.00	319.78	591.58	1082.46	848.87	580.54
v (m/s)	200.00	0.00	0.00	0.00	0.00	778.83

Table 1: Thermodynamic state summary

States: 1 inlet; 2 diffuser outlet; 3 compressor outlet; 4 combustor outlet; 5 turbine outlet; 6 nozzle exit. Zero velocities at states 2–5 indicate neglected internal kinetic-energy terms, not zero mass flow.

Model assumptions

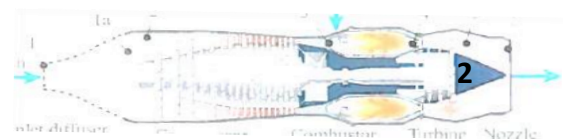
The flow is steady, with negligible potential-energy changes. The diffuser, compressor, turbine and nozzle are adiabatic and isentropic; the lecturer confirmed unit component efficiencies. The combustor is adiabatic at constant pressure, the shaft is lossless and hydrogen enters at the selected reference temperature of 298.15 K. Air is 21% O₂ and 79% N₂ by mole. Combustion is complete, nitrogen is inert, and product composition is frozen downstream.

NASA properties and verification

Species properties are combined by mass fraction: $h(T) = \sum Y_i h_i(T)$, $s(T) = \sum Y_i S_{\text{NASA}_i}(T)$, and $R_{\text{mix}} = R_u/M_{\text{mix}}$. The main script inverts monotonic property curves on a 200–3000 K grid with 1 K spacing. It uses air properties at states 1–3 and product properties at states 4–6. No constant- c_p Poisson relation or LHV heat-addition approximation is used.

All 21 conservation/property residual checks pass when the NASA functions are evaluated directly at the solved temperatures. The whole-engine energy residual is 3.289 W; the maximum enthalpy inversion error is 0.027605 J/kg (allowance 1 J/kg). The largest residual-to-tolerance ratio is 0.08214, below the pass limit of 1. The separate direct-root calculation agrees within 0.000446 K, 3.326 Pa and 0.000663 m/s. These checks establish numerical consistency of the assumed model.

Reproduction: extract 10_scripts.zip, set MATLAB Current Folder to the extracted folder and run verify_group10. This runs the model and reference calculation and exports the results. Verified with MATLAB R2026a.



Diffusor

```
128 v2=0;
129 h2=h1+0.5*v1^2-0.5*v2^2;
130
131 T2=invertProperty(hair_a,TR,h2,'Diffuser T2');
132 s2thermal=interp1(TR,sair_a,T2);
133
134 % S2=S1 determines the diffuser pressure rise.
135 P2=P1*exp((s2thermal-s1thermal)/Rg);
136 S2=s2thermal-Rg*log(P2/Pref);
137
138 diffuserEnergyResidual=(h1+v1^2/2)-(h2+v2^2/2);
```

Code snippet diffusor [1–2], JetEngine_Group10.m lines 128–138

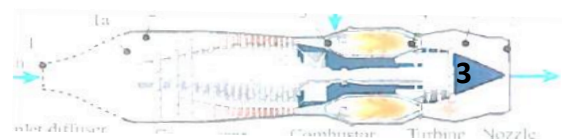
Procedure

For the steady adiabatic diffuser, shaft work and potential-energy changes vanish. The inlet velocity is 200 m/s and outlet kinetic energy is neglected. The energy balance therefore gives $h_2 = h_1 + v_1^2/2$ (line 129). The NASA air-enthalpy curve is inverted for T2 (line 131).

At unchanged composition, $s_2 = s_1$ gives $s_T(T_2) - s_T(T_1) - R_{air} \ln(P_2/P_1) = 0$. Thus $P_2 = P_1 \exp[(s_T(T_2) - s_T(T_1))/R_{air}]$, implemented at line 135. The thermal entropy s_T is the mass-weighted S_{Nasa} value. The fixed-composition mixing contribution cancels in the entropy difference.

The air mole fractions are converted to mass fractions using the database molecular masses (lines 75 and 79–81). The air enthalpy and entropy curves are assembled at lines 93–104. The inversion helper checks that its target lies within the available property range before interpolation.

The calculated diffuser outlet is $T_2 = 319.78$ K and $P_2 = 125.11$ kPa. The temperature and pressure rise are consistent with deceleration of the incoming air. Direct property evaluation gives a diffuser energy residual of 0.009484 J/kg and an entropy residual of 0.0008214 J/(kg K), below tolerances of 2 J/kg and 0.01 J/(kg K), respectively.



Compressor

```
151 P3=P3overP2*P2;
152 s3sthermal=s2sthermal+Rg*log(P3/P2);
153
154 T3s=invertProperty(sair_a,TR,s3sthermal,'Compressor T3s');
155 h3s=interpl(TR,hair_a,T3s);
156 S3s=s3sthermal-Rg*log(P3/Pref);
157
158 % eta_c=(h3s-h2)/(h3-h2)
159 h3=h2+(h3s-h2)/eta_c;
160 T3=invertProperty(hair_a,TR,h3,'Compressor T3');
161 s3thermal=interpl(TR,sair_a,T3);
162 S3=s3thermal-Rg*log(P3/Pref);
163
164 wcomp=h3-h2;           % Specific compressor work input [J/kg]
165 Wcomp=mair*wcomp;      % Compressor power input [W]
```

Code snippet Compressor [2–3], JetEngine_Group10.m lines 151–165

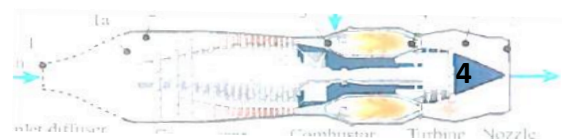
Procedure

The prescribed compressor pressure ratio gives $P_3 = 9P_2$. For the isentropic reference state, $s_{3s} = s_2$, so $sT_{3s} = sT_2 + R_{air} \ln(P_3/P_2)$. Line 154 inverts the NASA air-entropy curve to determine T_{3s} ; line 155 evaluates the corresponding enthalpy.

The efficiency definition is $\eta_c = (h_{3s} - h_2)/(h_3 - h_2)$. With $\eta_c = 1$, $h_3 = h_{3s}$ and $T_3 = T_{3s}$. Lines 159–162 retain the general efficiency relation and evaluate the outlet state using the air mixture.

The steady-flow energy balance gives compressor input power $\dot{W}_c = \dot{m}_a(h_3 - h_2)$, where $\dot{m}_a = AF \times \dot{m}_f = 204.42 \times 0.58 = 118.5636$ kg/s. Lines 164–165 give a specific work input of 280.371 kJ/kg and compressor power of 33.241777 MW.

The outlet is $T_3 = 591.58$ K and $P_3 = 1125.98$ kPa. Both rise relative to state 2. Direct NASA evaluation reconstructs the specified efficiency of 1 and gives a reference entropy residual of -0.0004973 J/(kg K). This small interpolation residual is within tolerance; it is not a physical entropy decrease.



Combustor (2 code snippets)

```

192 mprod=mair+mfurate;
193 mdot_in=mair*Yair+mfurate*Yfuel;
194 ndot_in=mdot_in./Mi;
195
196 nu=[-1 -0.5 0 1 0]; % Change per mole H2 burned
197 ndot_H2_burned=ndot_in(1);
198 ndot_prod=ndot_in+nu*ndot_H2_burned;
199 mdot_prod=ndot_prod.*Mi;
200
201 % Stoichiometric air-fuel ratio and equivalence ratio.
202 AF_st=(-nu(2)/Xair(2))*MAir/Mi(1);
203 phi=AF_st/AF;
204
205 %% Initial and final mixture composition
206 % Initial = air + fuel before combustion. Final = combustion products.
207 Yreac=mdot_in/sum(mdot_in);
208 Xreac=ndot_in/sum(ndot_in);
209 MReac=Xreac*Mi';
210 Rreac=Runiv/MReac;
211
212 Yprod=mdot_prod/sum(mdot_prod);
213 Xprod=ndot_prod/sum(ndot_prod);
214 MProd=Xprod*Mi';
215 Rprod=Runiv/MProd;

```

Composition before and after combustor [3–4], lines 192–215

```

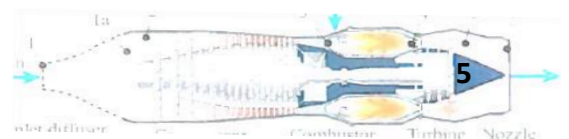
250 hprod_a=Yprod*hia';
251 sproda_a=Yprod*sia';
252
253 %% Combustor 3 -> 4
254 % Steady, adiabatic, no shaft work and negligible kinetic/potential energy.
255 % mair*h3+mfurate*hfuel=mprod*h4
256 % NASA enthalpies include formation enthalpy, so LHV is not added here.
257 hfuel=HNasa(Tfuel,SpS(1));
258 Hdot_in=mair*h3+mfurate*hfuel;
259 h4=Hdot_in/mprod;
260
261 T4=invertProperty(hprod_a,TR,h4,'Combustor T4');
262
263 % Combustion at constant pressure.
264 P4=P3;
265 s4thermal=interp1(TR,sproda_a,T4);
266 S4=s4thermal-Rprod*log(P4/Pref);

```

Thermodynamic computations [3–4], lines 250–266

Procedure

The species order is [H2, O2, CO2, H2O, N2]. Mass flows are converted to molar flows using molecular masses M_i . For $\text{H}_2 + \frac{1}{2} \text{O}_2 \rightarrow \text{H}_2\text{O}$, the reaction vector is $[-1, -0.5, 0, 1, 0]$. Applying it to the incoming H2 molar flow gives product molar flows; normalization gives X_{prod} and Y_{prod} . The pressure and thermal calculation are explained with Table 2 on the following page.



Combustor composition and procedure

AF = 204.42 (equivalence ratio = 0.16669)		
Mass fractions	Initial	Final
Fuel	0.004868	0.000000
O ₂	0.231784	0.193148
N ₂	0.763348	0.763348
CO ₂	0.000000	0.000000
H ₂ O	0.000000	0.043503
R _g [J/kg/K]	306.868	296.829
Table 2: Mixture composition before and after the combustor and the specific gas constants.		

Initial denotes the combined, unburned air-plus-fuel mixture; final denotes the combustion products. It does not mean air alone at state 3. Fractions are rounded for display; the unrounded mass and mole fractions each sum to 1.

Procedure

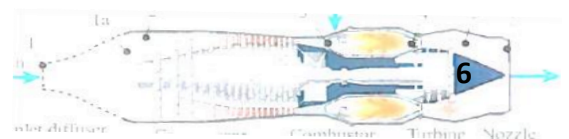
The stoichiometric air-to-fuel ratio is $AF_{st} = 34.0741$ kg/kg, giving $\phi = AF_{st}/AF = 0.16669$. The mixture is lean, leaving excess oxygen. Nitrogen passes through unchanged, and there is no carbon in the fuel, so no CO₂ is formed. Total product flow is $\dot{m}_p = \dot{m}_a + \dot{m}_f = 119.1436$ kg/s. Mass and H/O/C/N atom balances close to numerical precision.

For the steady adiabatic combustor, shaft work and kinetic- and potential-energy changes are negligible. The correct reacting-flow balance is $\dot{m}_a h_3 + \dot{m}_f h_f = \dot{m}_p h_4$. The fuel enthalpy is evaluated with H_{Nasa} at 298.15 K (line 257); lines 258–261 determine h_4 and invert the product-enthalpy curve for T₄. NASA enthalpies include formation enthalpy, so adding an LHV term would double-count chemical energy.

The results are T₄ = 1082.46 K and P₄ = P₃ = 1125.98 kPa. Although $h_4 \approx 300.807$ kJ/kg is slightly below $h_3 \approx 302.278$ kJ/kg, the temperature rises because both composition and mass flow change. The directly evaluated combustor energy residual is –0.3066 W. All subsequent calculations use $R_{prod} = 296.829$ J/(kg K) and the product property curves.

Sources

4EB00 Lecture 1 Ideal Gas Mixtures 2026, PDF pp. 19–27 (NASA formation enthalpies and mixture properties); Lecture 2 Cycle analysis 2026, PDF pp. 7–13 (control volumes and entropy method); supplied NASA database/functions. Report structure follows the 4EB00 Q1 2026 template; checks cover the component methodology, composition and results required by the rubric.



Turbine

```
327 v4=0;v5=0;
328 Wturb=Wcomp;
329
330 h5=h4-Wturb/mprod;
331 T5=invertProperty(hprod_a,TR,h5,'Turbine T5');
332
333 % eta_t=(h4-h5)/(h4-h5s)
334 h5s=h4-(h4-h5)/eta_t;
335 T5s=invertProperty(hprod_a,TR,h5s,'Turbine T5s');
336 s5sthermal=interp1(TR,sprod_a,T5s);
337
338 % S5s=S4 gives the turbine outlet pressure.
339 P5=P4*exp((s5sthermal-s4thermal)/Rprod);
340 S5s=s5sthermal-Rprod*log(P5/Pref);
341
342 s5thermal=interp1(TR,sprod_a,T5);
343 S5=s5thermal-Rprod*log(P5/Pref);
```

Code snippet Turbine [4–5], JetEngine_Group10.m lines 327–343

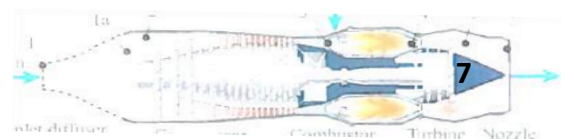
Procedure

The turbine drives only the compressor through a lossless shaft, so $\dot{W}_t = \dot{W}_c$. It processes $\dot{m}_p = 119.1436$ kg/s, whereas the compressor processes $\dot{m}_a = 118.5636$ kg/s. The balance is $\dot{m}_p(h_4 - h_5) = \dot{m}_a(h_3 - h_2)$, giving $h_5 = h_4 - \dot{W}_c/\dot{m}_p$ (line 330). Inverting the product-enthalpy curve yields T_5 .

The turbine efficiency is $\eta_t = (h_4 - h_5)/(h_4 - h_{5s})$. Lines 334–336 calculate the ideal reference state. With $\eta_t = 1$, $T_{5s} = T_5$. Its isentropic relation $s_{5s} = s_4$ gives $P_5 = P_4 \exp[(s_{T,5s} - s_{T,4})/R_{\text{prod}}]$, implemented at line 339. Product properties and the product gas constant must be used throughout.

The turbine outlet is $T_5 = 848.87$ K and $P_5 = 423.74$ kPa, with turbine power 33.241777 MW. Both temperature and pressure decrease, while the supplied power equals the compressor requirement. Using separately evaluated NASA enthalpies gives a shaft-balance residual of 1.018 W, below the 475.414 W tolerance derived from the state enthalpy allowances.

The exported state table also includes full mixture entropy using species partial pressures. For fixed product composition the mixing term is constant and cancels between states 4 and 5. It therefore does not change the turbine pressure calculation.



Nozzle

```
371 P6=Pamb;
372 assert(P5>P6,'Nozzle: P5 must be above ambient pressure. ');
373
374 % Ideal nozzle state: S6s=S5.
375 s6sthermal=s5thermal+Rprod*log(P6/P5);
376 T6s=invertProperty(sprod_a,TR,s6sthermal,'Nozzle T6s');
377 h6s=interp1(TR,hprod_a,T6s);
378 S6s=s6sthermal-Rprod*log(P6/Pref);
379
380 % eta_n=(h5-h6)/(h5-h6s)
381 h6=h5-eta_n*(h5-h6s);
382 T6=invertProperty(hprod_a,TR,h6,'Nozzle T6');
383 s6thermal=interp1(TR,sprod_a,T6);
384 S6=s6thermal-Rprod*log(P6/Pref);
385
386 % h5+v5^2/2=h6+v6^2/2
387 v6s=sqrt(v5^2+2*(h5-h6s));
388 v6=sqrt(v5^2+2*(h5-h6));
```

Code snippet Nozzle [5–6], JetEngine_Group10.m lines 371–388

Procedure

The exit pressure is prescribed as ambient: $P_6 = 100$ kPa. For the isentropic reference, $s_{6s} = s_5$ gives $sT_{6s} = sT_5 + R_{\text{prod}} \ln(P_6/P_5)$. Lines 375–378 invert the product-entropy curve and evaluate h_{6s} . With $\eta_n = (h_5 - h_6)/(h_5 - h_{6s}) = 1$, $h_6 = h_{6s}$ and $T_6 = T_{6s}$.

The adiabatic nozzle has no shaft work. Its energy balance is $h_5 + v_5^2/2 = h_6 + v_6^2/2$. With v_5 neglected, $v_6 = \sqrt{2(h_5 - h_6)}$, using enthalpies in J/kg. The results are $T_6 = 580.54$ K and $v_6 = 778.83$ m/s. The enthalpy drop is approximately 303.286 kJ/kg; the negative $h_6 = -281.485$ kJ/kg is permitted by the NASA formation-enthalpy reference.

The direct nozzle energy residual is 0.01220 J/kg, below its 2 J/kg tolerance. A local frozen-composition sound-speed calculation gives exit Mach number 1.6023. Thus the fully expanded solution assumes suitable nozzle geometry; it does not establish compatibility with an arbitrary converging-only nozzle or calculate shocks.

The integrated model therefore closes its mass, energy and reference-entropy balances and reproduces the independent numerical inversion within tolerance. Unit component efficiencies describe the ideal component limit, not 100% overall engine efficiency. Combustor entropy generation cannot be inferred from $s_4 - s_3$ alone because fuel enters separately and the mass flows differ.

